Candidate Guidelines

Dyad Technologies | QA Automation Practical Task

1. Problem Statement

You are asked to design and build automated test scripts for the scenario described in this document. It gives you the business context, the objective, a phased step-by-step workflow, and the validation criteria your automation must satisfy. You are expected to demonstrate a working, runnable automation project — not just a script that passes once by chance.

2. What We Expect From You

- Build an automated test project using any framework that is comfortable to you.
- Follow the Page Object Model (POM) design pattern to separate page interactions from test logic.
- Use explicit/fluent waits or framework auto-waiting; do not use hardcoded Thread.sleep() to work around timing issues.
- Handle dynamic data programmatically (e.g., dynamically identifying the cheapest flight, the highest/lowest priced product, or a generated employee ID) rather than hardcoding expected values.
- Implement every validation listed in the scenario's 'Validation Criteria' section as a real assertion in your code — not just a print/log statement.
- Include structured logging and/or a test execution report (e.g., TestNG/Extent/Allure report).
- Write clean, modular, readable code with meaningful naming, and remove commented-out/dead code before submission.
- Include a README.md describing the framework used, prerequisites, how to configure and run the tests, and any assumptions you made about the scenario.

3. Use of AI Tools

You are welcome to use any AI tool (Claude, ChatGPT, Copilot, Cursor, or similar) to help design your framework and write code — we're evaluating the working solution and your engineering judgment, not whether you typed every character yourself. However, you must be able to explain and defend your submission: be ready for your interviewer to point to any line of code, any file, or any design choice in your repository and ask you to walk through what it does and why it's written that way. Submissions the candidate cannot explain will be scored down regardless of whether the tests pass, so make sure you genuinely understand every part of what you submit before the review.

4. Timeline

This is a take-home assessment. You will have 2–3 days from the time the scenario(s) are assigned to you to submit your completed work. If you anticipate needing more time, reach out to your point of contact before the deadline rather than submitting late without notice.

5. GitHub Submission Process

- Create a GitHub repository for this assessment.
- Commit your work incrementally with clear, meaningful commit messages that reflect real progress (e.g., "Add POM for login page", "Implement cheapest-flight validation") — a single final commit dumping all the code is discouraged, since your commit history is part of the evaluation.
- Push all commits to the repository before the deadline.
- Ensure the repository is public (or grant access if asked to keep it private) so it can be reviewed without extra setup.
- Share the repository link with your interviewer / point of contact once you are done.
- Do not commit sensitive data (real credit card numbers, personal credentials, API keys) — use the dummy/test data the demo sites provide.

6. How You Will Be Evaluated

Criteria	What We Look For
Functional Correctness	All listed workflow steps and validation criteria are automated and pass reliably.
Framework & Design	Sensible use of POM, reusable utilities, and a maintainable project structure.
Handling of Dynamic Data	Correctly identifies dynamic values (prices, IDs, cheapest options) instead of hardcoding them.
Code Quality	Readable, consistent naming, no dead/commented-out code, sensible error handling.
Reporting & Logging	Clear execution reports/logs that make failures easy to diagnose.
Git Hygiene	Frequent, meaningful commits that show incremental progress, not one bulk commit.
Documentation	A README that lets someone else clone the repo and run the tests without guesswork.
Code Understanding	Candidate can clearly explain any line, file, or design decision when asked — regardless of whether AI tools were used to help write it.

Scenario 1: E-Commerce Order Validation

Site: https://www.saucedemo.com/

Candidate Name: _____ *Assessment Date:* _____

Business Scenario

A customer wants to purchase the most expensive and the least expensive products available on the store, and expects the order to be billed correctly and confirmed with a downloadable proof of purchase.

Objective

Automate the complete purchase journey — login, dynamic price-based product selection, checkout, and PDF invoice validation.

Workflow / Step-by-Step Task

Phase	Step #	Action / Instruction
Step 1: Login & Price Discovery	1	Login to the application.
Step 1: Login & Price Discovery	2	Fetch all product prices from the inventory page.
Step 2: Cart Building & Verification	3	Add the highest-priced and lowest-priced products to the cart.
Step 2: Cart Building & Verification	4	Verify cart total.
Step 3: Checkout	5	Complete checkout.
Step 3: Checkout	6	Validate order completion.
Step 4: Proof of Purchase	7	Generate PDF and verify correct product and Bill amount in PDF file.

Validation Criteria (Expected Outcomes)

#	Validation Criteria
1	Cart total equals the sum of the highest- and lowest-priced products.
2	Order completion is confirmed after checkout.
3	Generated PDF contains the correct product names and bill amount.